

Machine Learning for Natural Language Processing

Training Language Models to Follow Instructions



UNIVERSITY OF
GOTHENBURG

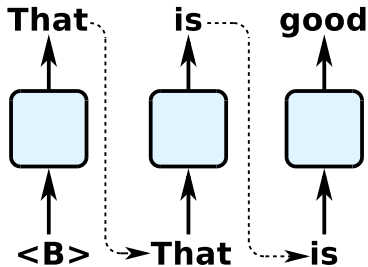
CHALMERS

Richard Johansson

`richajo@chalmers.se`

Starting point: Autoregressive text generation

$$\begin{aligned} P(x_1) \\ P(x_2|x_1) \\ P(x_3|x_1, x_2) \\ \dots \end{aligned}$$



Example

“Please write a poem about Python programming.”

Using gpt2-xl

```
input_text = 'Please write a poem about Python programming.'

input = tokenizer(input_text, return_tensors='pt').to('cuda')

outputs = model.generate(
    **input,
    max_length=100,
    do_sample=True,
    temperature=1.0
)

for i, output in enumerate(outputs):
    print(tokenizer.decode(output, skip_special_tokens=True))
```

Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
Please write a poem about Python programming.

Submit a topic about this contest. The best one is selected by the judges and published in the book.

Get some coffee, write some code, write more code, get on with it.

Questions? Contact us at pythonquest@gmail.com

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Zero-shot summarization with OPT-175B

Summarize the following document in one sentence: "Chalmers University of Technology (Swedish: Chalmers tekniska högskola, often shortened to Chalmers) is a Swedish university located in Gothenburg that focuses on research and education in technology, natural sciences, architecture, mathematics, maritime and other management areas. The university was founded in 1829 following a donation by William Chalmers, a director of the Swedish East India Company. He donated part of his fortune for the establishment of an "industrial school". Chalmers is one of only three universities in Sweden which are named after a person, the other two being Karolinska Institutet and Linnaeus University. Since 2012, Chalmers has achieved the highest reputation for Swedish Universities by the Kantar Sifo's Reputation Index. According to the survey, Chalmers is the most well-known university in Sweden regarded as a successful and competitive high-class institution with a large contribution to society and credibility in media."

I'm not sure what the point of this exercise is. I can't imagine anyone being able to summarize the above in one sentence

Supervised instruction fine-tuning

Write a Python one-liner that finds the longest string in a list of strings.

```
max(the_list, key=len)
```

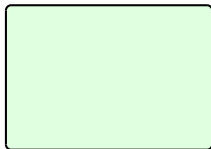
Supervised instruction fine-tuning

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```

$$P(\text{max(the_list, key=len)} \mid \text{Write a Python one-liner...})$$

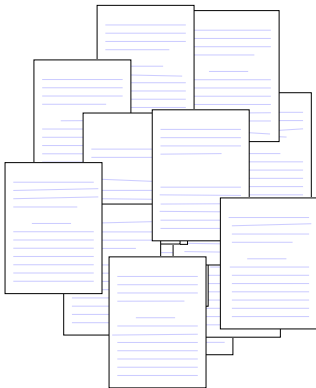
Pre-training $P(X)$



Instruction
fine-tuning



Massive collections of text



Large collections of
demonstrations

I am riding a bicycle. The pedals are moving fast, I look into the mirror and I am not moving. Why is this?

I am on a stationary bike.

Write me a funny poem about a cat driving a car.

I've seen a cat driving a car,
I saw it on my way to the bar,
I stopped to look and say hi,
I asked if he could give me a ride,
He looked at me and said no,
Then he drove off very slow,
I wonder where he was going,
Maybe he was just going bowling.

Example: Tasks in the Flan collection (Chung et al., 2022)

Commonsense
reasoning

I am riding a bicycle. The pedals are moving fast. I look into the mirror and I am not moving. Why is this?

I am on a stationary bike.

Generating code

Write a Python one-liner that finds the longest string in a list of strings.

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```

Is the following text toxic?
'I would like to order two vegetarian pizzas'

No

Toxicity classification

What was the name of Iggy Pop's band?

The Stooges

Closed-book
question answering

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I am riding a bicycle. The pedals are moving fast. I look into the mirror and I am not moving. Why is this?

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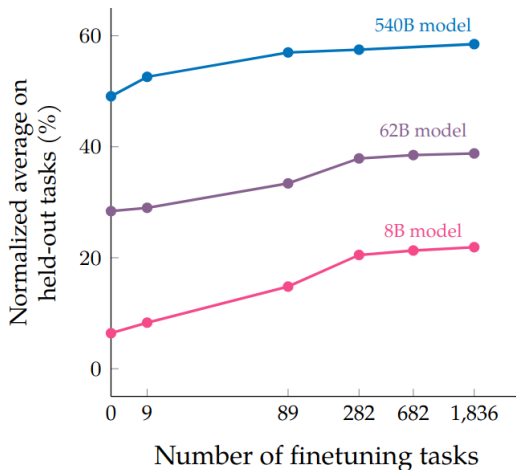
The Stooges

Closed-book
question answering

1836 tasks for **training**

102 tasks for **evaluation**

Adding more training tasks improves generalization



[image from (Chung et al., 2022)]

Various improvements

Paraphrasing instructions

Write a Python one-liner that finds the longest string in a list of strings.

Please give a one-line expression that can find the string in a given list that is the longest.

Step-by-step reasoning (Kojima et al., 2022)

What is the name of the capital of the second largest country in the world?

The second largest country in the world is Canada.

The capital of Canada is Ottawa.

Answer: Ottawa

The Flan Collection

[Mixtures](#) | [Download](#) | [Run](#) | [Flan Collection Paper](#) | [Citation](#)

The Flan Collection of datasets and data augmentation methods for instruction tuning is generated using the code in this repository. The Flan Collection compiles datasets from Flan 2021, [P3](#), [Super-Natural Instructions](#), along with dozens more datasets into one place, formats them into a mix of zero-shot, few-shot and chain-of-thought templates, then mixes these in proportions that are found to achieve strong results on held-out evaluation benchmarks, as reported for Flan-T5 and Flan-PaLM in the [Scaling Flan paper](#) and [Flan Collection paper](#).

List of Mixtures

We've broken down the Flan Collection into several sub-mixtures. These are "flan" (Flan 2021), "t0" (P3 excluding Flan 2021), "niv2" (Super-Natural Instructions) "cot" (several Chain-of-Thought datasets), and "dialog" (a few new dialog datasets). Each of these come in multiple varieties of templates: zero-shot prompts with answer options (zsopt), zero-shot prompts with no answer options (zsnoot), few-shot prompts with answer options (fsopt), and few-shot prompts with no answer options (fsnoot). Answer options indicate whether for multiple choice classification tasks the set of answers are described in the instruction prompt or not. These submixtures are instantiated in `flan/v2/mixtures.py`.

Download

If you'd like to directly download the final sub-mixtures, they are available here, courtesy of [Enrico Shippole](#):

- [Flan2021](#)
- [T0](#)
- [Niv2](#)
- [CoT](#)
- [Dialog](#)

<https://github.com/google-research/FLAN/tree/main/flan/v2>

(Longpre et al., 2023)

Using gpt2-xl-conversational

```
[51] input_text = '<|USER|> Please write a poem about Python programming. <|ASSISTANT|> '  
  
input = tokenizer(input_text, return_tensors='pt').to('cuda')  
  
outputs = model.generate(  
    **input,  
    max_length=100,  
    do_sample=True,  
    temperature=0.5, top_p=0.7, top_k=23, repetition_penalty=1.176,  
    pad_token_id=tokenizer.pad_token_id, eos_token_id=tokenizer.eos_token_id  
)  
  
for i, output in enumerate(outputs):  
    print(tokenizer.decode(output, skip_special_tokens=True))
```

Please write a poem about Python programming.

I've written a poem for you, sweetie:

"Python Programming":

The python is the way to do it,

and I'll teach you how to make it go

with your fingers and with my eyes.

And if you're ever in need of

a little help, just remember me!

My name's not really

Problems of supervised instruction fine-tuning

What is the name of the capital of Sweden?

Stockholm

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Write a Python one-liner that finds the longest string in a list of strings.

```
max(the_list, key=len)
```

```
max((len(s), s) for s in the_list)[1]
```

⋮

Problems of supervised instruction fine-tuning

What is the name of the capital of Sweden?

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```
max(the_list, key=len)
```

```
max((len(s), s) for s in the_list)[1]
```

⋮

Write a slogan about the values of Chalmers.

The core value of Chalmers is free expression.

We stand for inclusiveness, creativity, and sustainability.

⋮

Next: how to train the model with more high-level feedback

Reinforcement learning with human feedback

References

- H. Chung, L. Hou, S. Longpre, and B. Zoph et al. 2022. [Scaling instruction-finetuned language models](#). arXiv:2210.11416.
- T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa. 2022. [Large language models are zero-shot reasoners](#). In *ICML Workshop KRLM*.
- S. Longpre, L. Hou, T. Vu, and A. Webson et al. 2023. [The Flan collection: Designing data and methods for effective instruction tuning](#). arXiv:2301.13688.
- L. Ouyang, J. Wu, and X. Jiang et al. 2022. [Training language models to follow instructions with human feedback](#). arXiv:2203.02155.